HeartFailureTwin: End-to-End System Architecture

From Multimodal Waveforms & EHR to Patient-Specific Digital Twin Trajectory Forecasting

LAYER 1: DATA INGESTION & CLINICAL PROVENANCE

Historical EHR & Clinical

Demographics, Comorbidities,
Lab History, Prior HF (MIMIC/EHR)

Wearable ECG (500 Hz)

Lead-II Waveform, R-peaks,
RR intervals (PhysioNet SCG-RHC)

Tri-Axial SCG (500 Hz)

Chest Wall Seismocardiography,
Mechanics & Contractility Force

Wearable Hemodynamics

Photoplethysmography (PPG),
Respiration & Skin Temperature

LAYER 2: SIGNAL QUALITY AUDIT & PHYSIOLOGICAL PREPROCESSING

Clinical Profiler (ehr.py)

NYHA Stage, Charlson Index,
Hypertension, Diabetes Encoding

HRV & Peak Extractor (ecg.py)

Bandpass 0.5-40Hz, R-Peak Detection,
RMSSD, SDNN, Heart Rate (bpm)

Kinetics & Acoustics (wearable.py)

SCG RMS Energy, Dorso-ventral Force,
Spectral Centroid, Bandpass 1-40Hz

Quality Guard (quality.py)

Signal-to-Noise Ratio (SNR),
Clipping & Lead-Off [0.0 - 1.0]

LAYER 3: PATIENT-SPECIFIC BASELINE ENGINE (baseline.py)

Individual Baseline Calibration Envelopes: μ_i (rest), σ_i (dispersion)
Personalized Z-Score Deviations: $\Delta z_{i,t} = (x_{i,t} - \mu_i) / \sigma_i$ (Zero-Leakage)

LAYER 4: CROSS-MODAL GATED FUSION & LATENT TWIN STATE (state_estimator.py)

Non-Linear Gated Attention Fusion | Modality Embeddings d=16 | Quality-Weighted Gating
Latent State Space: $S(t) \in [-1, +1]^8$ (Normalized Physiological Twin State)

Trajectory Forecast (state_transition.py)

Neural ODE Transition S(t) -> S(t+1)
Horizon: 6h, 12h, 24h Decompensation Risk

In Silico What-If Engine (simulation.py)

Counterfactual Trajectory Simulation
(Diuretic Titration, Beta-Blockade, Stress)

Doctor Dashboard (React + FastAPI)

Cohort View, 4D Twin Replay (0-72h),
Early-Warning Radar & Transparent XAI